Ritwik Ala

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Summary

Computer Science undergraduate specializing in Data Analytics and AWS Certified Data Engineer – Associate, focused on building cloud-based data pipelines and analytical data platforms. Hands-on project experience with Python, SQL, PySpark, Amazon S3, Athena, Databricks, and Apache Airflow, including data transformation, dimensional modeling, and query-cost optimization. Seeking a Data Engineering Internship to build reliable, scalable data pipelines.

Education

VIT-AP University, B.Tech in Computer Science and Engineering (Data Analytics) 2023 – 2027
CGPA: **8.68/10**

Dr. KKR's Happy Valley School, Vijayawada 2021 – 2023
CBSE Class XII: **84.4%**

Dr. KKR's Gowtham School, Vijayawada 2021
CBSE Class X: **91.6%**

Technical Skills

Programming: Python, Java, SQL

Data Engineering & Big Data: PySpark, Apache Spark, Databricks, Apache Airflow, ETL/ELT Pipelines, Medallion Architecture, Batch Processing, Data Modeling (Star Schema, SCD Type 2), Data Cleaning & Transformation, Partitioning & Parquet Optimization

Databases: MySQL, Amazon Athena, SQL (Window Functions, Aggregations)

Cloud & AWS: Amazon S3, AWS Athena, AWS IAM

Data Visualization: Streamlit, Plotly, Matplotlib, Seaborn

Tools & Workflow: Git, REST APIs, Render

Projects

Algorithmic Trading & Quantitative Analytics Pipeline [Live Demo | GitHub Repo]

- Built a Python ETL pipeline using Pandas to process and transform **7,000+ historical stock market records** into a structured MySQL database.
- Designed relational database schema and SQL queries to compute key quantitative metrics including **Daily Returns** and **30-Day Volatility** for financial analysis.
- Developed an interactive Streamlit dashboard that automated the analysis of 7,000+ market records, enabling faster identification of market trends and data-driven insights.
- **Tech Stack:** Python, Pandas, MySQL, SQL, Streamlit, ETL, Data Visualization

Cloud E-Commerce Medallion Data Lakehouse [Live Demo | GitHub Repo]

- Architected a 3-tier Medallion (Bronze/Silver/Gold) pipeline ingesting **560K+** transactional e-commerce records and live REST API foreign exchange rates into Amazon S3.
- Developed distributed PySpark transformation jobs in Databricks to enforce strict schema rules, eliminate **10,000+** duplicates, and convert raw CSVs into date-partitioned Snappy Parquet, achieving a **79%** S3 storage footprint reduction and a **99%** Amazon Athena query scan cost reduction.
- Modeled curated Gold-layer data into a Star Schema featuring multi-currency fact tables (fact_orders), product dimensions, and SCD Type 2 history-tracked customer dimensions to capture evolving address changes.
- Orchestrated end-to-end daily incremental batch workflows utilizing Apache Airflow DAGs and least-privilege AWS IAM policies restricted strictly to project-specific S3 bucket resource actions.
- Designed and deployed an interactive Streamlit executive analytics application backed by Plotly visualizations, streaming curated Snappy Parquet tables from AWS S3 to track monthly revenue trends and customer repeat retention metrics.
- **Tech Stack:**Python, PySpark, Databricks, Amazon S3, AWS Athena, SQL, Streamlit, Plotly, Apache Airflow, ETL, Data Modeling

Certifications

- AWS Certified Data Engineer – Associate [View Badge]
- AWS Academy Graduate - Cloud Architecting [View Badge]
- AWS Academy Graduate - Cloud Foundations [View Badge]